Counterexample

When CEC fails, the counterexample is your shortest bug story , one input pattern where POs differ

The idea

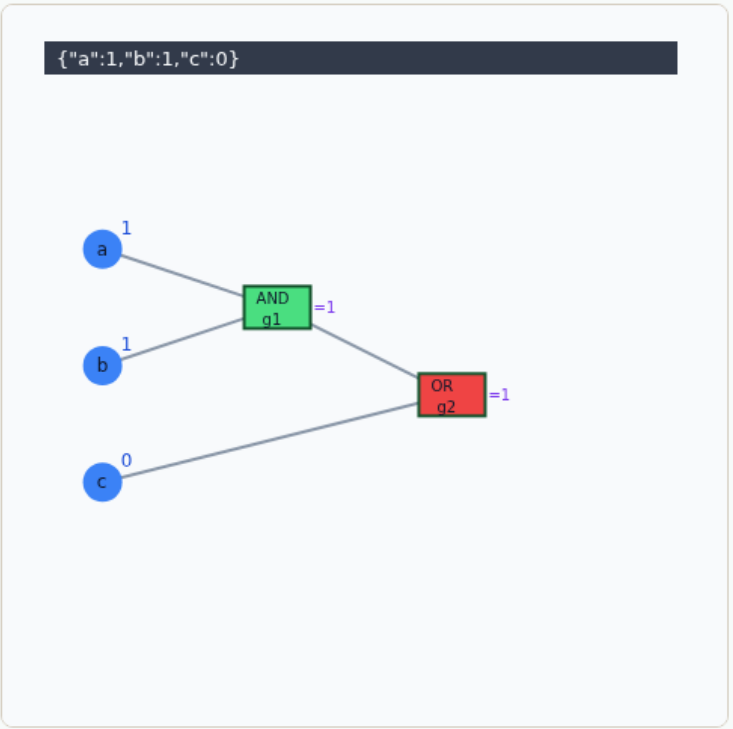
- Find_counterexample(spec, impl_bad) returns {'a':1,'b':1,'c':0}; spec_out=1, impl_out=0

Find cex

COUNTEREXAMPLE

Find cex

Step 1 / 5 · find · module02-07-counterexample



Explanation

find_counterexample(spec, bad) returns a=1,b=1,c=0.

- witness

```
GOLDEN
```

Find cex

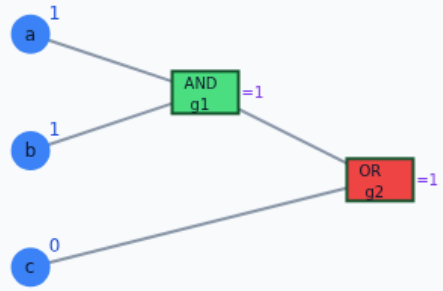
Simulate both

COUNTEREXAMPLE

Simulate both

Step 2 / 5 · sim · module02-07-counterexample

spec=1 bad=0



Explanation

At the cex, spec=1 and bad=0 — clear mismatch.

- debug

```
wave
```

Simulate both

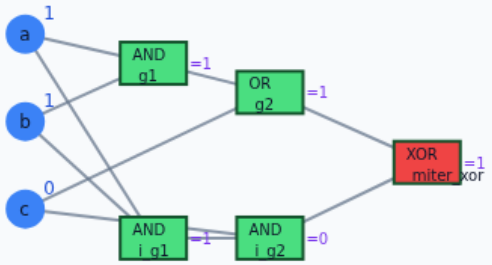
Miter lights

COUNTEREXAMPLE

Miter lights

Step 3 / 5 · miter · module02-07-counterexample

miter=1



Explanation

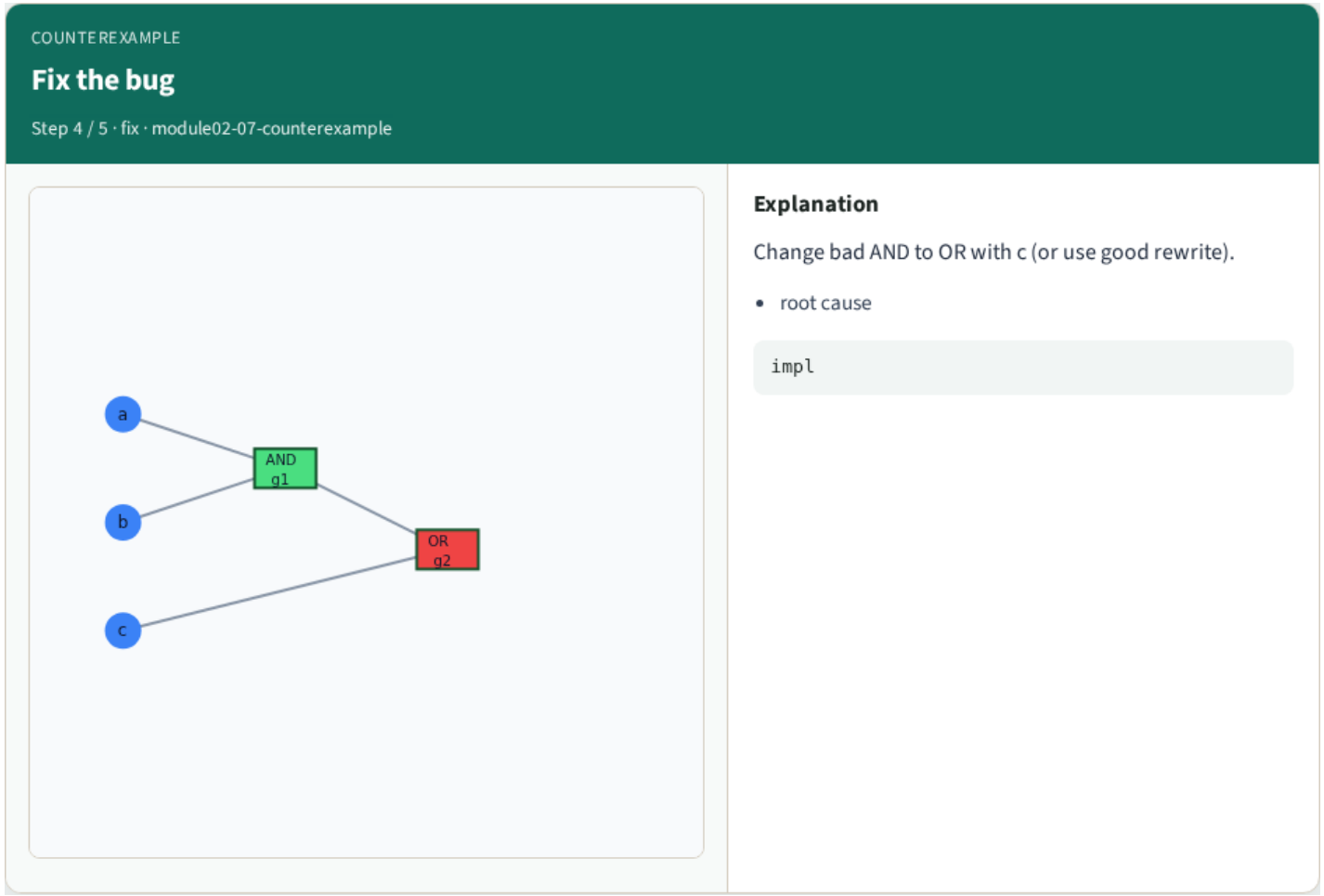
Bad miter at cex outputs 1.

- XOR=1

```
diff
```

Miter lights

Fix the bug



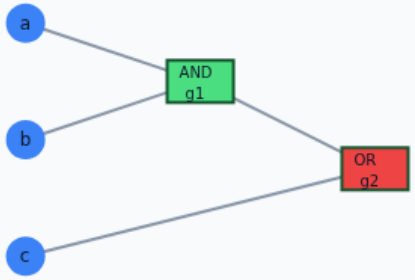
Fix the bug

Recheck CEC

COUNTEREXAMPLE

Recheck CEC

Step 5 / 5 · recheck · module02-07-counterexample



Explanation

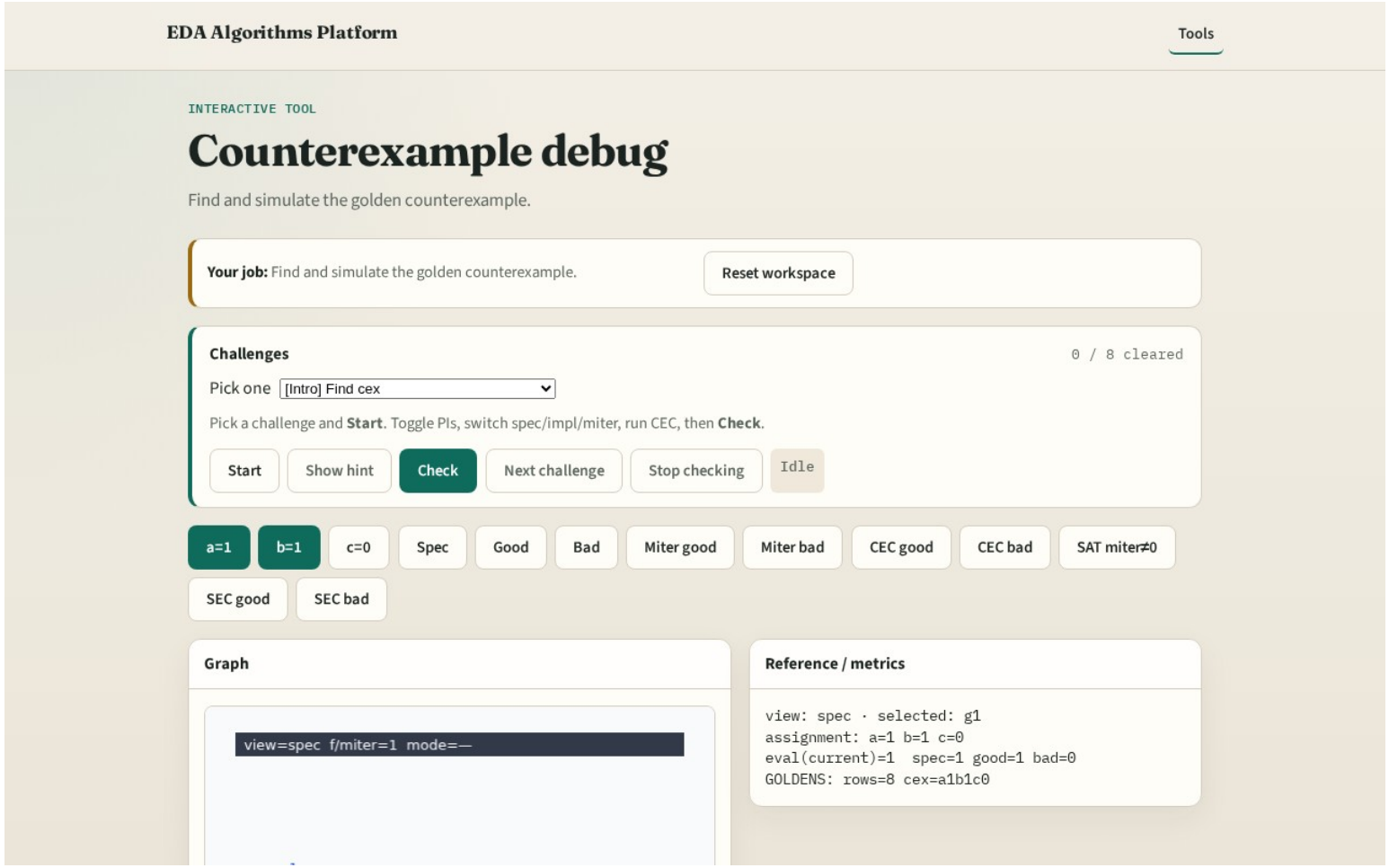
After fix, truth-table CEC should pass.

- regress

```
Next DC
```

Recheck CEC

Browser lab track



Browser lab starter

Implement track

- Call find_counterexample(spec, impl_bad); print spec and impl PO at cex

Pitfalls

- Ignoring cex and random sim
- Fixing only one corner without regression

Your turn

- Next: don't-care CEC